

Supplementary Material to “A Multidimensional Agent-Based Modeling and Simulation Framework for Urban Water Demand Forecast”

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This supplementary material support the manuscript with comprehensive documentation about structure of Agent-Based Model for Water Prediction (ABM-WP). Section 1 presents a summary of work focused on water demand forecasting, in comparison to ABM-WP, with approach used, dataset, contribution, and validation technique. Section 2 presents the description of the Performance Measures, Environment, Actuators, and Sensors (PEAS) for the three agents of the framework: Analyzer Agent (AA) in Table 2, Predictor Agent (PA) in Table 3, and Communicator Agent (CA) in Table 4.

Section 3 details the architectural project developed using the Tropos methodology, illustrated in Figures 1 and 2, which show the interaction between human actors (family, operator, meter, attendant, technician, meter) and the three agents. Section 4 characterizes the consumption profiles by sector in the Itapuã neighborhood, with Figure 3 showing the demarcation of the 102 sectors and the location of the Abaeté Metropolitan Park. Section 5 presents the comparative results of the 18 ABM-WP scenarios in relation to the SARIMA model (Figure 4), revealing that the scenarios combining population growth with income dynamics (CIV–CIX) show the largest positive discrepancies, while the environmental scenarios (CII, CX–CXII) show deviations close to zero.

Section 6 displays the monthly water consumption projections for all scenarios, along with the LSTM and SARIMA models (Figure 5), demonstrating the strategic advantage of ABM-WP in capturing demand trajectories driven by income inequality and behavioral change. Finally, Section 7 presents details of the Shapiro-Wilk normality test results in Table 5, and confirms the normal distribution for all ABM-WP scenarios ($p > 0.05$), unlike the SARIMA model, which rejected normality ($p = 0.0000$).

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1. Overview of Water Demand Forecasting Frameworks

Table 1 presents an overview of water demand forecasting frameworks organized by reference, approach, dataset location and scale, primary contribution, and validation techniques compared to ABM-WP.

Table 1: Overview focusing on ABM, categorized by solution approach, dataset location and scale, primary contribution, and validation technique. Source: Elaborated by the author.

Reference	Approach	Dataset (location & scale)	Primary Contribution	Validation technique
Galán et al. (2009)	ABM + GIS	Valladolid, Spain (2,000 agents)	Integration of spatial and social dynamics for domestic water management.	Statistics, Time Series
Koutiva and Makropoulos (2016)	ABMS	Athens, Greece (3,000 households)	Evaluation of the efficiency of management policies during drought periods.	Time Series
Sharvelle et al. (2017)	IUWM (GIS-based)	Fort Collins, USA (Block groups)	Assessment of demand management under climate and land use change scenarios.	Statistics, Time Series
Darbandsari et al. (2017)	ABMS	Tehran, Iran (3,172 households)	Simulation of behavioral characteristics and social interactions for demand management.	Statistics, Time Series
Rasoulkhani et al. (2018)	ABMS	Miami Beach, USA (380 households)	Analysis of the influence of income, pricing, and rebate policies on conservation technology adoption.	Statistics, Sensitivity
Searle and Harper (2020)	ABMS	Cape Town, South Africa (1,000 households)	Investigation of residential water conservation tendencies in response to demand management strategies.	Statistics, Time Series
Perelló-Moragues et al. (2021)	Socio-cognitive ABM	Barcelona, Spain (300 households)	Simulation of complex decision-making via a socio-cognitive architecture for urban demand.	Statistics, Expert Knowledge
Canales et al. (2024)	GW-ABM (Review)	Global (77 studies)	Identification of challenges in the simulation of groundwater and society.	-
Vidal-Lamolla et al. (2024)	ABMS	Barcelona, Spain (1,200 households)	Simulation of socio-cognitive agents for the assessment of policy implementation.	Expert Knowledge
Botero-Acosta et al. (2025)	SWAT (Hydro-logic)	Kaskaskia River, USA (Watershed)	Estimation of the isolated impacts of urban expansion on streamflow under climate change.	Statistics, Time Series
Ehsanitarbar et al. (2025)	ABMS	Tehran, Iran (5,000 households)	Management of reservoir demand via social and hydrological interactions.	Statistics, Expert Knowledge
ABM-WP framework	Spatially explicit ABMS (GAMA)	Salvador, Brazil (19,630 households)	Comparative analysis of ABM against statistical and ML approaches, regarding the influence of demographics and income dynamics.	Statistics, Time Series, Expert Knowledge

2. Performance Measures, Environment, Actuators, and Sensors (PEAS) to describe the task environment of the AA, PA, and CA agents.

Table 2: PEAS description for the AA, responsible for data collection, preprocessing, and analysis. Source: Elaborated by the author.

PEAS	Description
Performance	Quantity and quality of data processed per consumer unit
Environment	Virtual with georeferenced addresses. Classified as Non-deterministic, Dynamic (independent changes), Discrete, Partially observable, Sequential (history-dependent)
Actuators	Analyze family-grouped data Compare and process consumption Send processed data to predictor agent
Sensors	Residence geolocation Residence types and profile Population growth

Table 3: PEAS description for the PA, responsible for making scenario simulations. Source: Elaborated by the author.

PEAS	Description
Performance	Maintain prediction quality with minimal Mean Absolute Percentage Error (MAPE)
Environment	Virtual with georeferenced addresses. Classified as Non-deterministic, Dynamic (independent changes), Discrete, Partially observable, Sequential (history-dependent)
Actuators	Select optimal prediction model Generate family consumption forecasts Send forecasts to AC
Sensors	Data provided by AA

Table 4: PEAS description for the CA, responsible for consolidating the generated forecasts. Source: Elaborated by the author.

PEAS	Description
Performance	Number of reports with consumption forecasts issued
Environment	Virtual with georeferenced addresses. Classified as Non-deterministic, Dynamic (independent changes), Discrete, Partially observable, Sequential (history-dependent)
Actuators	Send water demand reports Distribute to PA water demand reports
Sensors	Data generated by the PA

3. Architectural Design

The architectural model of the ABM-WP framework was structured using the Tropos methodology, which details the interaction between human actors and the agents responsible for forecasting the water demand. The Operator is the central point of progression, managing data dependencies with the Family (household data provider) and the Meter (responsible for recording monthly and historical consumption). Operational intelligence is distributed across three specialized agents: the Analyzer Agent, responsible for analyzing, preprocessing, and consolidating customer data; the Predictor Agent, generating individualized consumption forecasts per family and per scenario; and the Communicator Agent, which prepares and consolidates scenario predictions. The workflow is driven by dependencies and objectives.

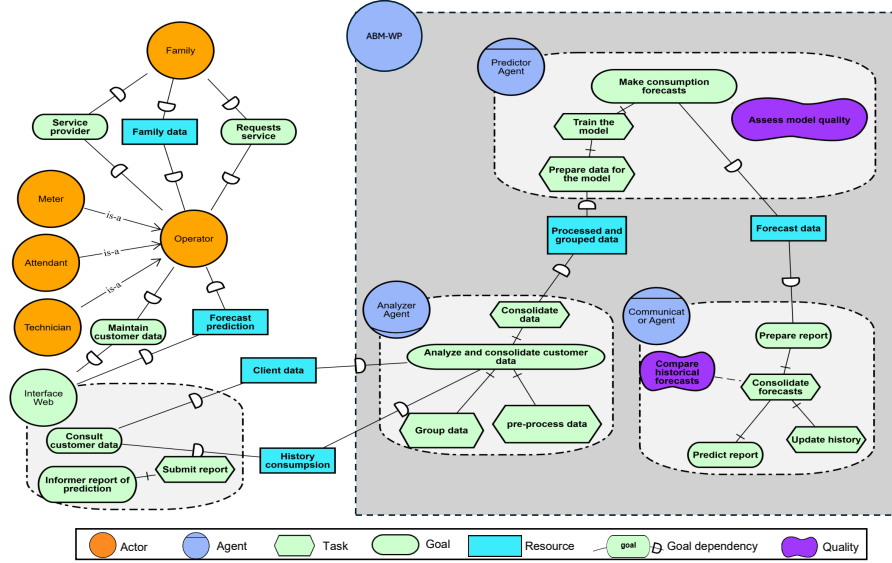


Figure 1: Architecture of the ABM-WP framework, encompassing the interaction between actors, tasks and agents.

Figure 2 presents the final requirements of the project, highlighting key of the Tropos elements (Tasks, Resources, Goals, and Dependencies), ensuring that each agent operates in a coordinated manner to achieve the main objective. The AA collects and processes historical data on consumption, type of residence, and family income, grouping them by consumer unit and preparing them for the next step. Then, the PA (one agent instantiated for each consumer unit) makes individualized future consumption forecasts. Each agent forecasts consumption for a household, taking into account its behavior profile and relevant variables. Finally, the CA consolidates the forecasts, compares them with historical data, and generates reports that are made available for use. This agent-based approach allows for a modular and scalable structure, where each

agent plays a specific role, ensuring accurate forecasts and providing more detailed information for efficient planning in water resource management.

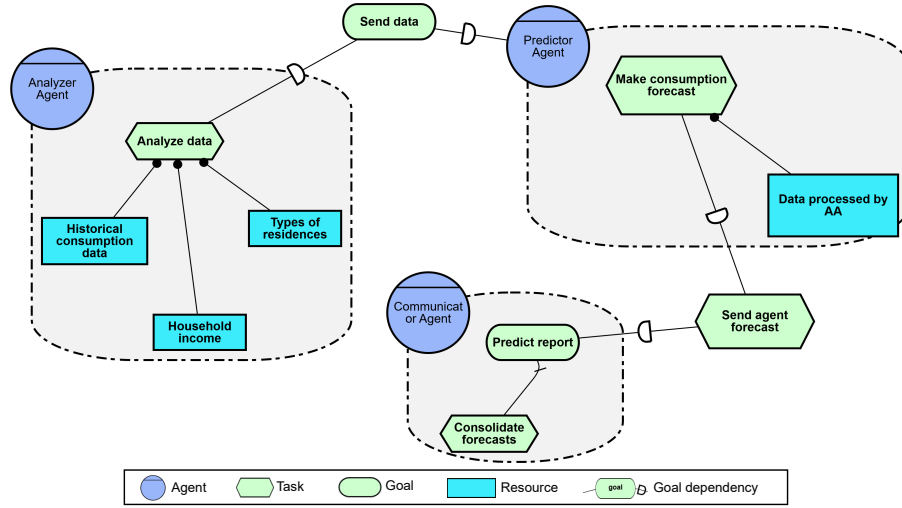


Figure 2: Tropos final requirements model with the AA, PA, and CA, using the piStar tool. Source: Elaborated by the author.

4. Consumption Profiles by Sector



Figure 3: Demarcation of the Itapuã neighborhood, separated by 102 sector limits in yellow, with the Abaeté Metropolitan Park visible on the right in green, located within an environmental preservation area. Active water connections are those represented by blue circles. Source: Elaborated by the author.

Ref.: April/2026

CD_SETOR	total_consumers	environmentalist_count	moderate_count	wasteful_count	avg_daily_consumption	pct_environmentalist	pct_moderate	pct_wasteful
S019	387	286	35	66	66.26	73.90	9.04	17.05
S096	376	258	16	102	93.32	68.62	4.26	27.13
S098	308	223	22	63	78.74	72.40	7.14	20.45
S007	343	211	18	114	152.81	61.52	5.25	33.24
S014	272	204	19	49	68.66	75.00	6.99	18.01
S057	339	203	23	113	126.51	59.88	6.78	33.33
S032	356	199	27	130	137.48	55.90	7.58	36.52
S054	338	199	41	98	108.89	58.88	12.13	28.99
S099	281	187	20	74	121.86	66.55	7.12	26.33
S002	331	174	26	131	122.85	52.57	7.85	39.58
S024	263	173	22	68	95.01	65.78	8.37	25.86
S039	209	168	15	26	53.14	80.38	7.18	12.44
S037	246	166	14	66	80.33	67.48	5.69	26.83
S042	238	161	11	66	84.35	67.65	4.62	27.73
S093	211	161	10	40	60.76	76.30	4.74	18.96
S033	195	160	16	19	49.99	82.05	8.21	9.74
S041	334	156	29	149	145.97	46.71	8.68	44.61
S005	288	156	22	110	166.17	54.17	7.64	38.19
S055	273	154	34	85	100.85	56.41	12.45	31.14
S030	244	151	22	71	107.47	61.89	9.02	29.10
S029	199	151	8	40	75.31	75.88	4.02	20.10
S083	244	149	26	69	103.62	61.07	10.66	28.28
S095	275	148	27	100	118.71	53.82	9.82	36.36
S068	258	146	29	83	131.13	56.59	11.24	32.17
S051	216	144	19	53	89.61	66.67	8.80	24.54
S026	247	143	20	84	106.38	57.89	8.10	34.01
S059	195	140	16	39	82.06	71.79	8.21	20.00
S015	227	139	14	74	103.99	61.23	6.17	32.60
S028	205	139	14	52	87.03	67.80	6.83	25.37
S092	175	139	10	26	58.11	79.43	5.71	14.86
S023	205	138	11	56	83.56	67.32	5.37	27.32
S074	221	135	18	68	91.53	61.09	8.14	30.77
S034	177	133	5	39	72.13	75.14	2.82	22.03
S067	194	132	14	48	89.87	68.04	7.22	24.74

Ref.: April/2026

CD_SETOR	total_consumers	environmentalist_count	moderate_count	wasteful_count	avg_daily_consumption	pct_environmentalist	pct_moderate	pct_wasteful
S020	261	129	19	113	128.31	49.43	7.28	43.30
S016	239	125	14	100	161.33	52.30	5.86	41.84
S062	140	121	6	13	38.99	86.43	4.29	9.29
S022	193	119	5	69	135.53	61.66	2.59	35.75
S040	167	118	15	34	75.77	70.66	8.98	20.36
S004	244	117	22	105	148.04	47.95	9.02	43.03
S061	162	116	10	36	75.32	71.60	6.17	22.22
S087	177	115	19	43	88.39	64.97	10.73	24.29
S003	221	113	13	95	178.36	51.13	5.88	42.99
S080	142	113	8	21	57.54	79.58	5.63	14.79
S075	145	108	8	29	61.84	74.48	5.52	20.00
S049	135	108	8	19	57.05	80.00	5.93	14.07
S011	246	106	17	123	218.82	43.09	6.91	50.00
S101	214	103	23	88	145.98	48.13	10.75	41.12
S082	171	101	22	48	105.64	59.06	12.87	28.07
S047	202	100	12	90	180.90	49.50	5.94	44.55
S094	217	98	22	97	167.44	45.16	10.14	44.70
S058	155	98	8	49	108.41	63.23	5.16	31.61
S053	139	97	8	34	87.06	69.78	5.76	24.46
S050	149	95	16	38	104.17	63.76	10.74	25.50
S010	166	92	14	60	149.82	55.42	8.43	36.14
S006	218	91	19	108	194.79	41.74	8.72	49.54
S086	121	91	9	21	68.83	75.21	7.44	17.36
S077	177	90	10	77	149.01	50.85	5.65	43.50
S090	136	90	9	37	95.20	66.18	6.62	27.21
S027	152	87	12	53	116.85	57.24	7.89	34.87
S031	257	86	21	150	173.21	33.46	8.17	58.37
S008	132	86	8	38	97.85	65.15	6.06	28.79
S025	188	84	10	94	173.88	44.68	5.32	50.00
S056	128	83	11	34	102.50	64.84	8.59	26.56
S078	104	83	7	14	63.40	79.81	6.73	13.46
S043	153	82	18	53	153.99	53.59	11.76	34.64
S046	139	81	12	46	112.37	58.27	8.63	33.09
S073	159	79	7	73	173.25	49.69	4.40	45.91

Ref.: April/2026

CD_SETOR	total_consumers	environmentalist_count	moderate_count	wasteful_count	avg_daily_consumption	pct_environmentalist	pct_moderate	pct_wasteful
S070	224	74	19	131	292.26	33.04	8.48	58.48
S076	163	74	14	75	153.64	45.40	8.59	46.01
S038	157	73	22	62	126.04	46.50	14.01	39.49
S069	223	72	26	125	244.18	32.29	11.66	56.05
S013	169	71	12	86	332.14	42.01	7.10	50.89
S089	99	70	5	24	82.39	70.71	5.05	24.24
S084	164	69	19	76	140.89	42.07	11.59	46.34
S100	94	69	7	18	313.97	73.40	7.45	19.15
S063	71	64	1	6	34.08	90.14	1.41	8.45
S009	134	63	9	62	157.06	47.01	6.72	46.27
S088	109	62	11	36	127.88	56.88	10.09	33.03
S071	146	61	16	69	144.48	41.78	10.96	47.26
S045	122	59	10	53	197.04	48.36	8.20	43.44
S060	66	59	1	6	30.27	89.39	1.52	9.09
S079	85	57	10	18	81.72	67.06	11.76	21.18
S021	77	55	3	19	101.33	71.43	3.90	24.68
S035	252	49	28	175	182.38	19.44	11.11	69.44
S072	93	49	5	39	303.72	52.69	5.38	41.94
S044	88	49	4	35	122.07	55.68	4.55	39.77
S064	117	48	9	60	166.09	41.03	7.69	51.28
S065	98	44	12	42	133.51	44.90	12.24	42.86
S018	86	43	8	35	237.17	50.00	9.30	40.70
S066	59	40	6	13	77.25	67.80	10.17	22.03
S048	93	35	9	49	227.41	37.63	9.68	52.69
S102	128	31	10	87	380.86	24.22	7.81	67.97
S012	117	30	10	77	273.30	25.64	8.55	65.81
S052	110	29	6	75	431.88	26.36	5.45	68.18
S103	135	28	15	92	300.14	20.74	11.11	68.15
S036	242	27	19	196	207.61	11.16	7.85	80.99
S091	35	14	1	20	163.97	40.00	2.86	57.14
S097	2	1	0	1	162.57	50.00	0.00	50.00
S085	1	0	0	1	74360.66	0.00	0.00	100.00

5. Comparative results of ABM-WP scenarios vs. SARIMA model

Figure 4 shows the annual discrepancies between each of the 18 ABM scenarios and the SARIMA statistical forecast over the 2026–2035 horizon. Positive values indicate that the ABM scenarios projects consumption above the SARIMA baseline, while negative values indicate lower consumption. The scenarios combining population growth with income dynamics (CIV–CIX) exhibit the largest positive discrepancies, reaching close to $100,000 \text{ m}^3$ by 2035, with the income imbalance scenarios (CVI and CIX) showing the largest deviations. The specific waste scenarios (CIII and CXIII–CXV) also exhibit substantial positive discrepancies, though slightly smaller than those in the population-income scenarios. In contrast, the environmentally focused scenarios (CII and CX–CXII) show discrepancies close to zero or slightly negative over the projection period. The random income scenarios (CXVI–CXVIII) exhibit intermediate behavior, with discrepancies ranging from 20,000 to 40,000 m^3 . These results demonstrate that population growth and income dynamics, and not just behavioral profiles, are primarily responsible for the divergence of the ABM-WP model from statistical predictions.

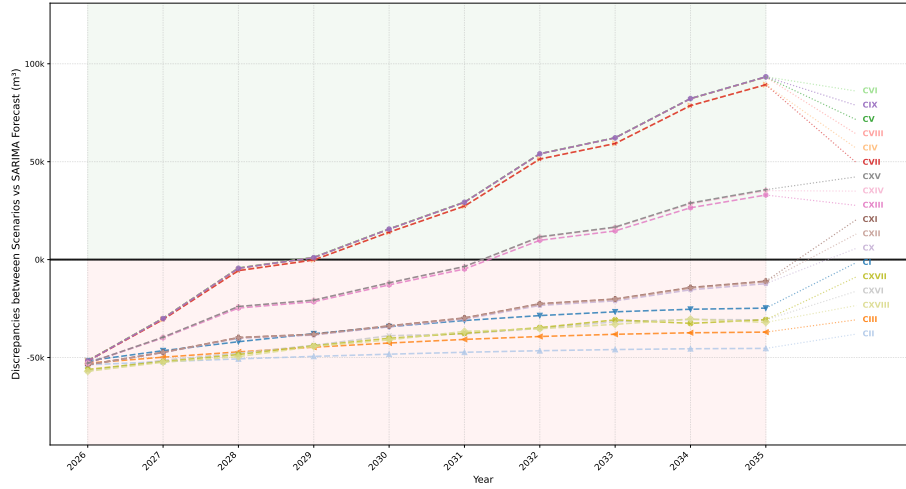


Figure 4: Annual discrepancies between ABM scenarios and SARIMA forecast (2026–2035). Positive values indicate higher consumption projected by ABM scenarios. Scenarios combining population growth with income dynamics (CIV–CIX) show the largest positive discrepancies. Environmentalist scenarios (CII, CX–CXII) show near-zero deviations.

6. Projection of ABM-WP scenarios vs LSTM and SARIMA model

Figure 5 presents monthly water consumption projections for all 18 ABM scenarios, alongside LSTM neural network and SARIMA statistical forecasts over the 2026–2035 horizon. While SARIMA and LSTM offer precision in capturing recurring seasonal patterns, ABM-WP offers a strategic advantage: even without fine-tuning for seasonality, it identifies demand trajectories driven by income inequality and behavioral change, something essential for long-term planning in socioeconomic transition scenarios.

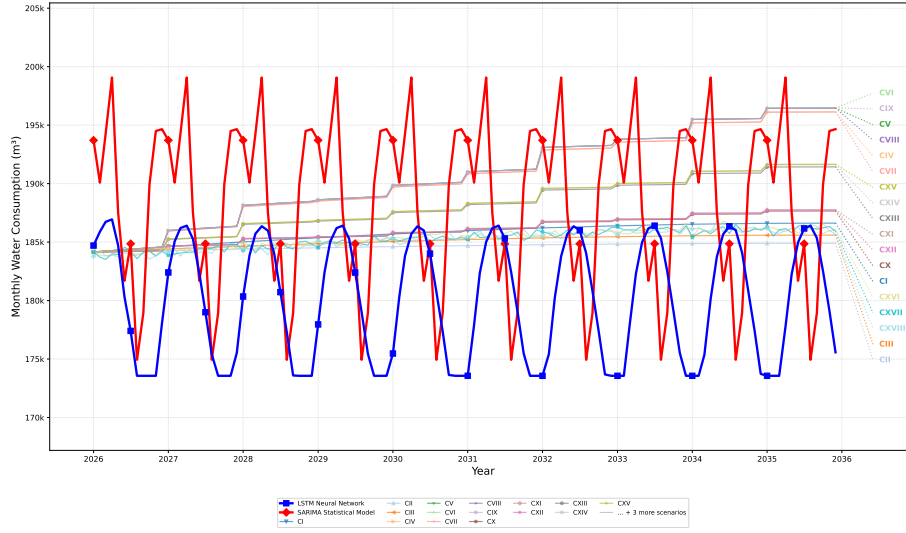


Figure 5: Monthly water consumption projections (2026–2035) for 18 ABM scenarios, LSTM neural network, and SARIMA statistical model.

7. Shapiro-Wilk Normality Test

Table 5: Shapiro-Wilk normality test results for water demand scenarios showing the test statistic (W), p -values, and normality assessment for each consumption scenario. Source: Elaborated by the author.

Scenario	statistic (W)	p-value	Normal Distribution
CI	0.9173	0.3347	Yes
CII	0.9173	0.3352	Yes
CIII	0.9172	0.3344	Yes
CIV	0.9701	0.8915	Yes
CV	0.9702	0.8928	Yes
CVI	0.9703	0.8932	Yes
CVII	0.9701	0.8917	Yes
CVIII	0.9702	0.8930	Yes
CIX	0.9703	0.8933	Yes
CX	0.9700	0.8909	Yes
CXI	0.9701	0.8922	Yes
CXII	0.9701	0.8921	Yes
CXIII	0.9701	0.8917	Yes
CXIV	0.9702	0.8931	Yes
CXV	0.9703	0.8937	Yes
CXVI	0.9195	0.3531	Yes
CXVII	0.9198	0.3550	Yes
CXVIII	0.9234	0.3861	Yes
SARIMA	0.5685	0.0000	No

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